Vivek Vanera

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Education

North Carolina State University Aug 2025 – May 2027
Master of Computer Science *GPA: 4.0/4.0*
• Coursework: Design and Analysis of Algorithms, Software Engineering, Automated Learning and Data Analysis, Advanced Topics in Machine Learning

Institute of Technology, Nirma University Oct 2021 – May 2025
Bachelor of Technology in Computer Science and Engineering with Minor in Robotics *PPI: 8.63/10.0*
• Coursework: Machine Learning, Deep Learning, Natural Language Processing, Database Management Systems, Probability and Statistics, Graph Theory, Big Data Analysis, Software Engineering, Microservice Architecture and Programming

Work Experience

GenAI & Machine Learning Engineering Intern, *Seagate Technology* *May 2026 – Present*
- Built **topic classification** for an internal HR **RAG** assistant using example-driven prompts, scoping queries to the correct domain and improving retrieval relevance
- Integrated and tested **JWT authentication** for an AI intake agent across multiple config flows
- Evaluated retrieval strategies (**HyDE**, multi-query **RRF**, parent-child chunking, hybrid search) for a patent-search system using an **LLM-as-judge** framework
- Identified and escalated two **governance risks** in a production **Copilot** deployment, including an unmonitored privacy exposure

Data Science Intern, *RoutedAI, Inc.* *Jan 2025 – May 2025*
- Managed the re-design of the **job recommendation system**, reducing response time by **30%**
- Led the development of APIs for career transitions, delivering personalized insights and mapping similar career paths
- Developed an API to **detect the best hiring periods** of companies (e.g., quarterly peaks), helping users time applications strategically and maximize hiring opportunities
- Optimized the platform's resume parser, improving its parsability and reducing processing time by **20%**
- Built data-driven solutions utilizing **AWS**, **Supabase**, **Snowflake**, and **FastAPI**; integrated **LLMs** (OpenAI, DeepSeek, Groq, Mistral) to improve accuracy and interpretability

Projects

EZmail: AI-Powered Smart Email Agent
- Developed an **AI-powered email agent** that automatically classifies new emails by importance and extracts key details
- Implemented features like email summarization, link and date extraction, interactive cards with priority-based color coding, **reducing manual inbox monitoring**
- Integrated **automated response generation** with adjustable tone, allowing users to modify and send replies seamlessly
- Deployed as a **containerized FastAPI microservice**, enabling scalable and low-latency performance

Recofy: A RAG Based Context-Aware Music Recommendation System
- Built a music recommender system with **RAG**, leveraging **Spotify's API** for user authentication and metadata retrieval
- Generated song tags via **Last.fm's** and **OpenAI's** APIs, generating **vector embeddings** stored in **Pinecone DB** for similarity search
- Designed as a knowledge-grounded system, where retrieved data from Pinecone represents user's liked songs and playlists, ensuring contextually personalized recommendations
- Recommended songs by retrieving **top-3** matches from Pinecone DB **achieving 85%+** retrieval accuracy and refining suggestions through OpenAI's API

State-Machine Multi-Agent Refund Resolver
- Designed an orchestrator-owned **finite state machine** with **guarded transitions** and a **human-approval gate** for high-value refunds; agents can only propose actions, never skip or reorder workflow steps
- Built **four FastAPI agents** (classifier, policy, action, verifier) on strict **Pydantic contracts**, with **PostgreSQL** state persistence per transition, plus retries, timeouts, and **idempotency keys** that prevent duplicate refunds (tested across **N** scenarios with **0** duplicate refunds)
- Built a **hybrid decision layer** that uses **yes/no question-based judging (LAYA)** for fuzzy-text calls like intent and claim consistency, validated with **CheckEval**, cutting median end-to-end latency ∼**84%** vs. an LLM-only baseline

Skills

Programming Languages: Python, C/C++, Solidity, SQL, HTML, Lua
AI/ML Technologies: Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Audio Processing, GenAI, Data Pre-processing, Data Visualization, Time-series Analysis, PyTorch, Keras, Tensorflow, Large Language Models, Retrieval-Augmented Generation, Generative Adversarial Networks, Vector Databases, Embeddings, OCR
Databases: MongoDB, MySQL, Supabase, Snowflake, Pinecone DB, AWS S3
Tools: Google Cloud Platform, AWS Redshift, AWS Athena, Git/GitHub, Docker, Spark, Hadoop, FastAPI, n8n, JWT/Authentication